RESEARCH METHODOLOGY

Introduction to Research

Contents

- What is Research
- Scientific Inquiry

What is Research?

Session Objectives

- To understand the meaning of research
- To understand the meaning of research methodology
- To understand the objectives of research
- To understand the basic steps of conducting research

Session Outline

- Definitions:
  - research
  - methodology
  - research methodology
- The aims of research
- Basic steps to conduct research

What is research?

- Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested

More understanding of research

- Research is a careful and systematic process of inquiry to find answers to **problem** of interest.
- Research includes: use of **facts**, use of **theories**, data(fact) **analysis**, sampling, doing an experiment, going to library to read up on a topic
- To do research is to **investigate the problem** systematically and thoroughly
- Goal of research is to solve '**problem**' of interest

Research characteristics

1. Originates with a question or problem.
2. Requires clear articulation of a goal.
3. Follows a specific plan or procedure.
4. Often divides main problem into sub-problems.
5. Guided by specific problem, question, or hypothesis
6. Accepts certain critical assumptions.
7. Requires collection and interpretation of data.
8. Cyclical(helical) in nature.

Aims of research

- To describe about a phenomena
- To predict about ‘something’ in the future
- To explain about solution of problem
- To interpret what is being investigated

Scientific methods

- “Methodos” means way
- Methodology is the discipline of scientific procedures

What is methodology?

- A formalized approach to conduct research
  - A series of steps and deliverables
- Research methodology in computer science/Information System/Information Technology

“ A collection of methods, procedures, techniques, tools and documentation aids which help the researcher in their efforts to conduct a research in the field of CS/IS/IT”.

Research Methodology Vs. Research Method

- ***Research methodology:***
  - conceptual frameworks and assumptions used to inform research
  - Elaboration of research method
- ***Research method:***
  - Research technique or procedure used to gather and analyses data

Research methodology

- Research methodology will consist of phases, sub-phases, which will guide the researchers in their choice of methods, techniques, procedures, tools, etc., that might be appropriate at each stage of the research and also help them plan, manage, control and evaluate the progress of research.
- A research methodology represents a way to develop research systematically.
- A research methodology represents a way to do scientific inquiry.
- A research methodology should have a sound theoretical basis.

Why methodology

- It will make you better able to understand and utilize scientific information in both your personal life and your work.
- It will make you more literate and cultured person.
- It provides you some insight not only into particular scientific findings but also into the general nature of science as a human activity.

Nature of research in CS/IS/IT

- Research in CS often requires experimental design as well as theoretical(basic) research ---
  - Hence, we have to have a methodology in order to conduct research CS
  - ***Science approach***: new knowledge, theory
- Research in IS/IT often requires system development
  - Hence, we have to have a methodology in order to develop a system (IS/IT system)
  - ***Engineering approach***: construction of useful products
- This IS/IT development methodology can be used as a research methodology in IS/IT field.

How to start a research

- Determine a well defined problem:
  - Understanding about something
    - ***Science approach:*** phenomena, theory, hypothesis, experiment.
  - Constructing product
    - ***Engineering approach:*** build a system that performs a (unique) task, experiment, convergence results (stable)..
- Follow the steps in the research methodology
- Research results: knowledge, algorithm, methods, products(system), model, etc…

Problem Formulation

Problem Formulation

- **Identification of research problem**
- **Identification of scope of research**
- **Identification of research questions**
- **Identification of research objectives**

→ Hypothesis

Model of Scientific Inquiry

Session objectives

- To understand the steps in scientific inquiry
- To understand the characteristics of Scientific Methods
- To understand the logical thinking in Research
- To understand various types of Research

Session Outline

- What is Scientific Inquiry?
- Model of Scientific Inquiry
- Inductive and Deductive Logical Thinking
- Types of Research methods

What is Scientific Inquiry?

- Scientific Inquiry is a term that *encompasses a variety of techniques* that scientists use *to explore the natural world* and propose explanations based on the evidence they find.
- The objective of scientific inquiry is *to find and to characterize the patterns* as resulted from the exploration.
- Scientific inquiry is founded on *experiment and observations* as opposed to purely rational or isolated logical thought
- Scientific inquiry can enrich our understanding of science

Why Scientific Inquiry?

- It will make you more knowledgeable about something.
- You will be able to explain about something more thoroughly
- It will make you a more literate and cultured person.
- It will provide you more ideas

Characteristics of Scientific Inquiry

- Based on facts
- Objective Consideration
- Analytical
- Quantitative in Nature
- Deductive Logic-Hypothesis
- Inductive Logical-Generalization

Activities in Scientific Inquiry

- Making observations
- Posing questions
- Finding out is already known
- Planning investigations
- Reviewing past knowledge in light of experimental evidence
- Using tools to gather, analyze, and interpret data
- Proposing explanations
- Communicating results

Models of Scientific Inquiry

- Classical model
  - The classical model of scientific inquiry derives from Aristotle, who distinguished the forms of approximate and exact reasoning, set out the threefold scheme of abductive, deductive, and inductive interference, and also treated the compound forms such as reasoning by analogy.
- Pragmatic model
  - Chales Pierce considered scientific inquiry to be a species of the genus inquiry, which he defined as any means of fixing belief, that is , any means of arriving at a settled opinion on a matter in question.
  - He observed that inquiry in general begins with a state of uncertainty and moves toward a state of certainty, sufficient at least to terminate the inquiry for the time being.

General model of scientific inquiry

- State general problem
- Conduct Literature Search
- State Specific problem
- Design methodology
- Gather data
- Analyze data
- Report results
- Refine hypothesis

Another Model

- Problem statement
- Organize Critical Thinking Framework
- Organize premises
- Hypothesis formulation
- Testing hypothesis
- Draw conclusions

Output of Each step

- State general problem
  - Statement that stated general problems
  - Example: will the universe expand for ever?
    - Exercise: State at least three general problems on your own.
- Conduct Literature Search
  - List all related literatures to the problems stated
  - Underline all theories, techniques, methods, findings and so forth that others have found
  - Critically analyze all the weaknesses, strengths, similarities, dissimilarities among those theories, techniques, findings above.
  - Summarize all those literatures (make annotated bibliographies)

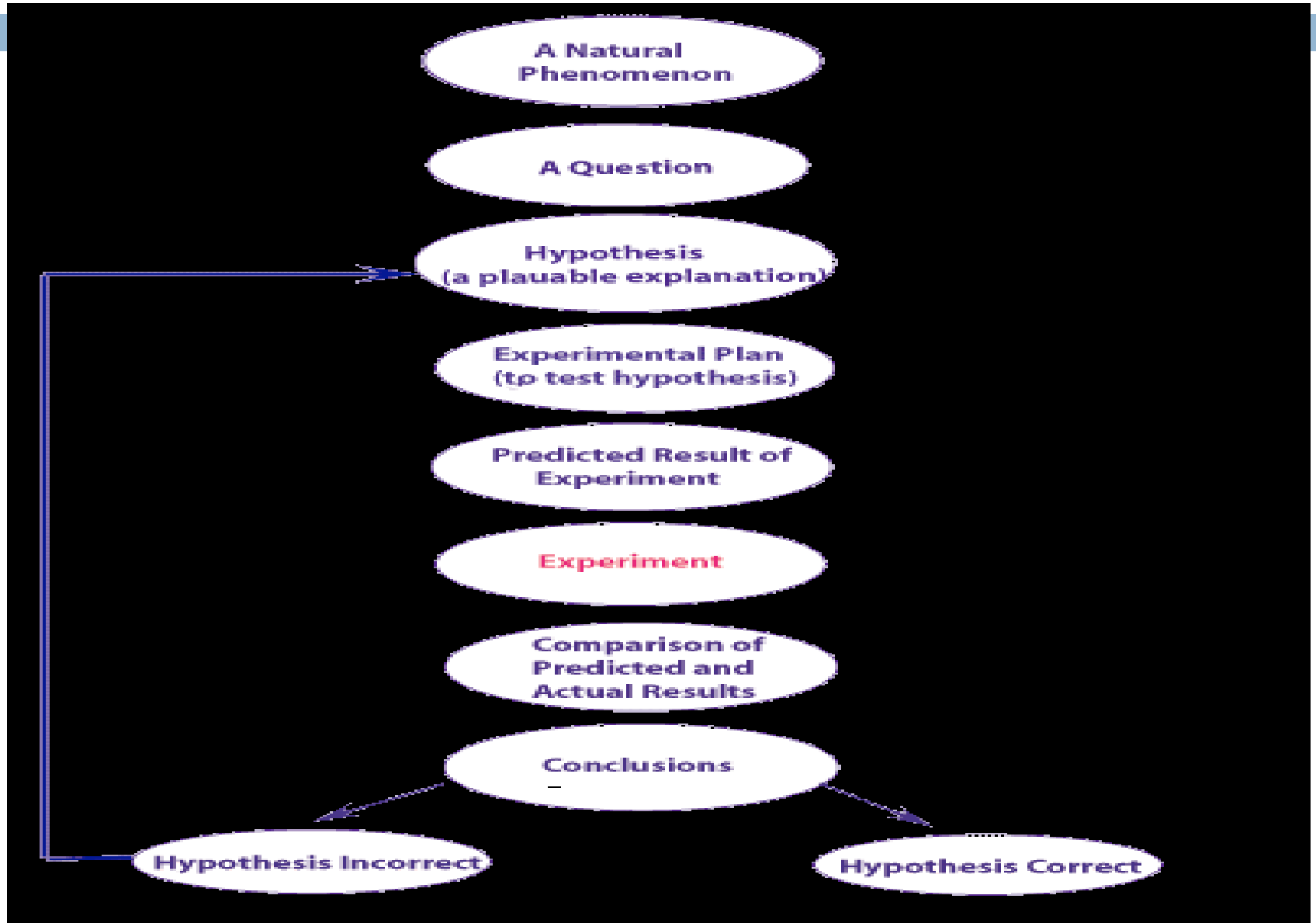
Output of Each Step

- State specific problem
  - Statement of general problem to specific problem
  - Example: What are the factors that cause the universe expanded?
    - Exercise: State at least two specific problems that derived from the above example
- Design Methodology
  - List all the steps to be taken in order to answer those stated problems
  - Establish the setting of experiment
  - Provide all necessary means, tools, instruments
  - Define the object, variables, etc

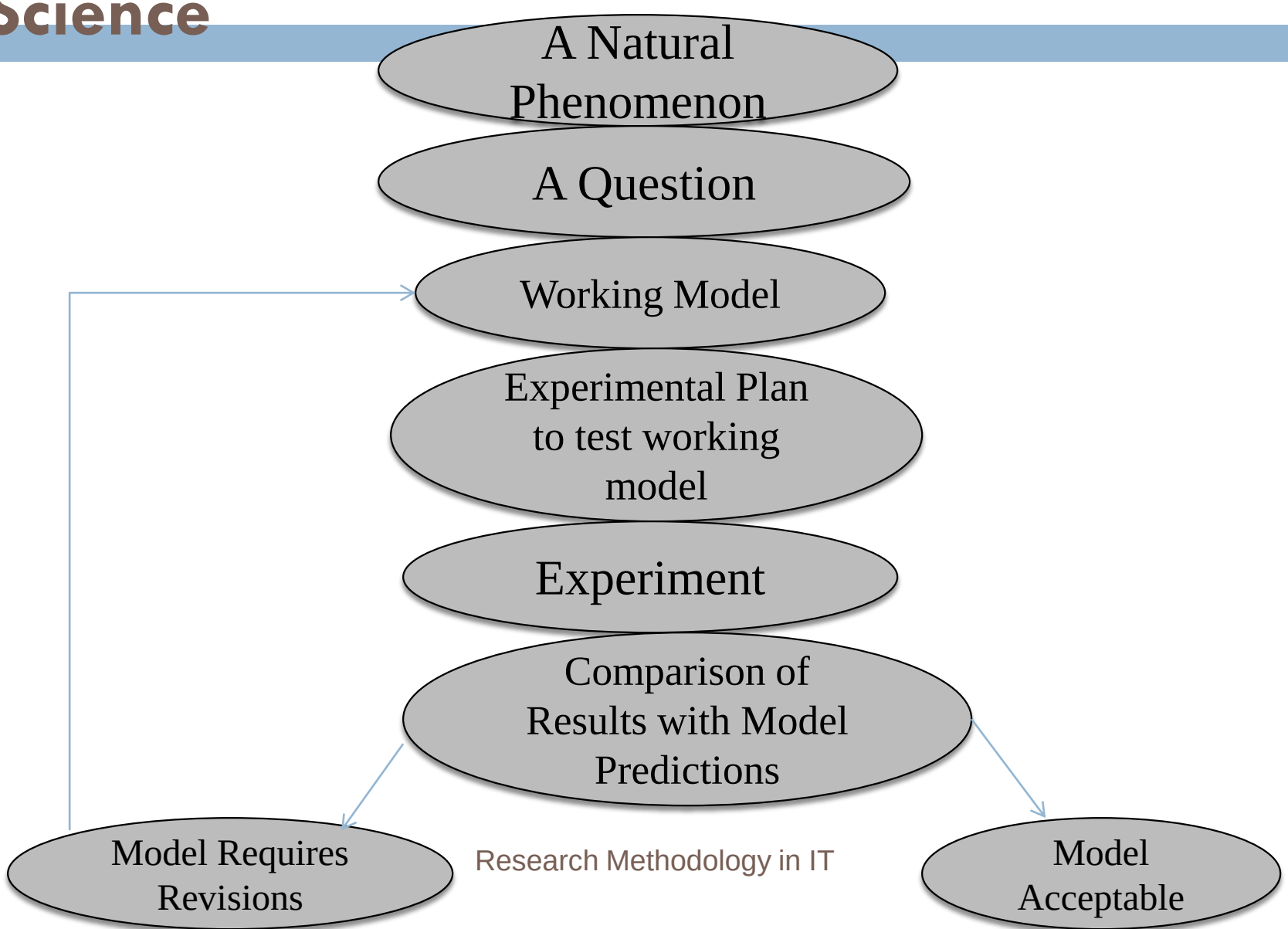
Output of Each Step

- Gather Data
  - Organize data in tables, graphics
- Analyze Data
  - Interpret the organized, and processed data
- Report the Results

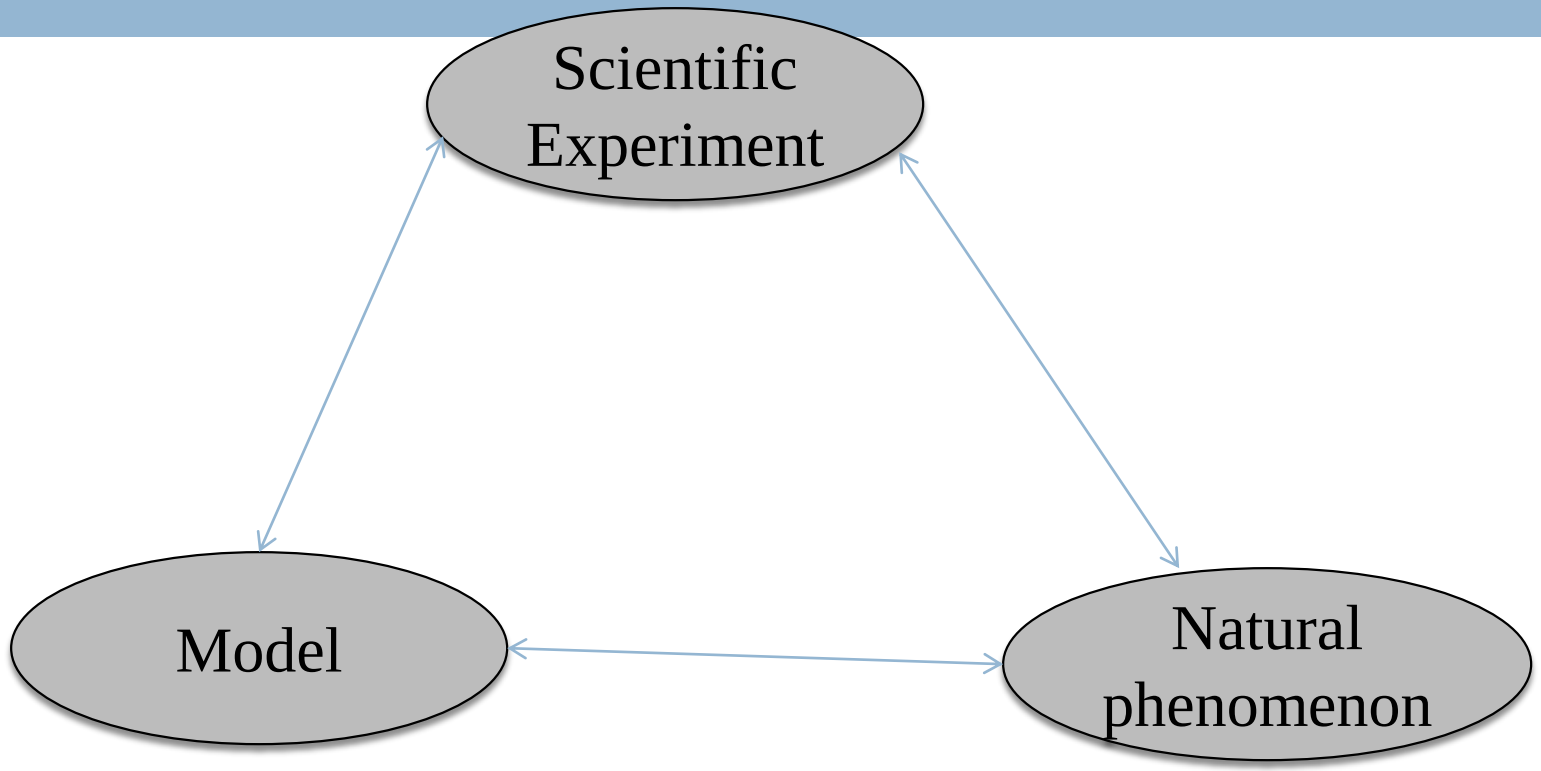
Simplest Form of Scientific Inquiry



Model Development in the Physical Science



Key Components of Scientific Inquiry



Example of Model Development

What Factors Determine the Visibility of a Rainbow?

Working Model

Sunlight travels straight through rain drops where it is dispersed into colors, and then the light travels straight into my eyes

Proposed Experiment

While it is raining early one morning, I will face toward the rising sun in the ease to view a rainbow.

Predicted Results

I will see a rainbow in the sky as I look east on a rainy morning

Actual Results

No rainbow was observed toward the east. However , I did see rainbow when I turned around and looked west

Conclusion

My working model requires revision. The sun must be behind me to view a rainbow. There must be another effect in addition to light being dispersed into color in the rain

Basic Skills of Scientific Inquiry

- Observing
- Classifying and sequencing
- Communicating
- Measuring
- Predicting
- Hypothesizing
- Inferring
- Defining, controlling, and manipulating variable in experimentation
- Designing, constructing and interpreting models
- Interpreting, analyzing and evaluating data

Uses of Scientific Inquiry

- Make connections with world situations.
- To encourage more active problem solving approach to learning and thinking
- To apply math skills
- Review what is already known in light of experimental evidence
- Proposes answers, explanations and predictions
- Use tools to gather, analyze and interpret data

Exercise

- Develop a scientific model of your own
  - post deliverables of each step of the model
- Categorize the scientific model into a research method
- Give examples of deductive and inductive logics